

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 08:33 UTC

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Generated 2026-05-21 08:33 UTC

- Cycles recorded: **1001**
- Time span: 658.9h (~1.52 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	955
BARRIER_HIT	26
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	59
Schur-Weyl Duality	36
Free Probability	26
Representation Theory of Symmetric Groups	24
Matroid Theory	23
Free Probability Theory	19
Additive Combinatorics	15
Noncommutative L^p Geometry	12
Random Matrix Theory	11
Spectral Graph Theory	11
Noncommutative Harmonic Analysis	10
Polymatroid Theory	9
REAL_ALGEBRAIC_GEOMETRY	9
Algebraic Geometry	8
Persistent Homology	8
Algebraic Topology	7
Finite Geometry	6
Schur-Weyl duality	6
Plethysm Theory	6
Invariant Theory	6
Fourier Analysis on Boolean Functions	5
Additive combinatorics	5

Field	Cycles
Algebraic Geometry of Secant Varieties	5
FOURIER_ANALYSIS	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Non-Abelian Harmonic Analysis	4
COMMUNICATION_COMPLEXITY	4

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-21 03:30 UTC	INCONCLUSIVE	Laplacian Eigenvalue and Resolution Length of Tseitin Formulas
2026-05-21 03:38 UTC	INCONCLUSIVE	Moment Matrix Spectral Gap and SOS Degree for Max-CUT
2026-05-21 03:52 UTC	INCONCLUSIVE	Free Entropy Gap in Read-Twice BP Transition Matrices
2026-05-21 04:29 UTC	INCONCLUSIVE	Matroid Rank Gap in Monotone DNF for k-CLIQUE
2026-05-21 04:56 UTC	INCONCLUSIVE	Algebraic Shifting Facet Count Bounds Tseitin Resolution Length
2026-05-21 05:31 UTC	INCONCLUSIVE	Symmetric Group Orbit Count Distinguishes Read-Twice BPs
2026-05-21 05:41 UTC	INCONCLUSIVE	Polynomial Threshold Function Degree Bounds
2026-05-21 06:02 UTC	INCONCLUSIVE	Karchmer-Wigderson Pr Fourier Coefficient Sum
2026-05-21 06:31 UTC	INCONCLUSIVE	Bounds Monotone DNF Size for k-CLIQUE
2026-05-21 06:56 UTC	INCONCLUSIVE	Real Radical Dimension Bounds AC^0 Circuit Size for PARITY
2026-05-21 07:37 UTC	INCONCLUSIVE	Eigenvalue Count and SOS Degree for Max-CUT Approximation
2026-05-21 07:51 UTC	INCONCLUSIVE	Prime Density in Arithmetic Progressions and Seed Length of Nisan
2026-05-21 08:09 UTC	INCONCLUSIVE	Free Entropy Gap in Disjointness Communication Matrices
2026-05-21 08:09 UTC	INCONCLUSIVE	SOS Moment Matrix Spectral Gap and Max-CUT Approximation Ratio

Time	Verdict	Title
2026-05-21 08:19 UTC	INCONCLUSIVE	Additive Energy of Fourier Coefficients and ACC ⁰ Circuit Size
2026-05-21 08:33 UTC	INCONCLUSIVE	Betti Number Sum Lower Bound for SOS Max-CUT Degree

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```